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Carboxylic Acids and Derivatives

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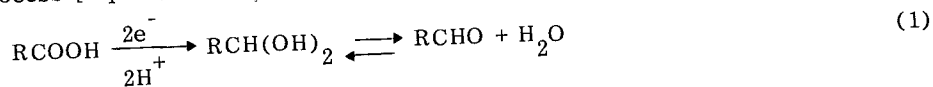
I. INTRODUCTION

This chapter is concerned with the cathodic reduction of carboxylic acids and their derivatives, i.e., esters, amides, anhydrides, hydrazides, and nitriles. Cyclic derivatives of substituted carboxylic and polycarboxylic acids, such as lactones, lactams, imides, and anhydrides, will also be included. Only those transformations in which the functional group itself is involved are treated.

Useful cathodic conversions of the carboxyl group and its derivatives have been long known and tabulations and discussions of these reactions can be found in several places [1-5]. Most of these processes involve constant-current reduction with careful selection and control of solvent, temperature, pH, and electrode material. Only recently have mechanistic studies of these processes been made and our knowledge in this area is far from complete.

II. CATHODIC REDUCTION OF CARBOXYLIC ACIDS

The cathodic reduction of a carboxyl group can give either the corresponding aldehyde in a two-electron process or the alcohol in an overall four-electron process [Eqs. (1) and (2)]:

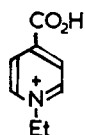


It is difficult to stop the reduction at the aldehyde stage since aldehydes generally are reduced at less negative potentials than carboxylic acids. This transformation is possible in certain cases, some of which will be discussed below.

Aliphatic and aromatic acids which are not strongly activated by electron-withdrawing groups and do not contain more easily reducible groups give a well-developed polarographic wave in aqueous [6] or nonaqueous [7,8] solvents, but this is usually due to reduction of protons to hydrogen. As an example, the hydrogen wave from acetic acid in acetonitrile/0.1 M tetraethylammonium perchlorate appears at -2.3 V (SCE) and in aqueous medium in the intermediate pH region at -1.76 V (SCE). Controlled potential electrolysis (CPE) of acetic acid in acetonitrile at -2.3 V (SCE) demonstrates that the electrode process actually is the reduction of protons. No second wave is found in the accessible potential region of these media, indicating that RCOO^- , the species left after reduction of the proton, is very difficult to reduce. Unless activated, therefore, it is unlikely that carboxylic acids will be easily reduced in slightly acidic or alkaline solution or in nonaqueous media. In strongly acidic medium the situation may be different since protonation of the carboxyl group might lead to the formation of a species capable of accepting electrons into the carbon-to-oxygen bonds.

For reduction of dicarboxylic acids, in acetone solution, polarographic half-wave potentials, and limiting currents, are dependent on concentration in a way which suggests that at low concentration the acids are sufficiently dissociated for reduction of solvated protons to predominate, whereas at higher concentrations the more difficult reduction of undissociated acid is observed [9].

Acids containing certain activating groups can be reduced without interference from hydrogen evolution. 1-Ethyl-4-carboxypyridinium ion [(I), as the bromide] is such a case [10]; the carboxyl group is the most easily reducible group at pH values below 7. The half-wave potential of (I) moves toward more negative potentials with increasing pH in such a way that the $E_{1/2}$ -pH curve can be approximated by three lines: the first one (pH 0.5-1.5) with a slope of 0.029 V/pH, the second one (pH 2-7) with a slope of



(I)

0.068 V/pH, and the third one (pH 8-13) with a slope of 0.007 V/pH. The total variation of $E_{1/2}$ is between -0.74 V in strongly acidic and -1.22 V in alkaline solution. Hydrogen evolution is at yet more negative potentials. From this pH dependence it was concluded that from pH 8 to 13 the reducible species contains the carboxyl group as the carboxylate ion and below 7 existed in its undissociated form. Since the pH of the acid is 1.75, the carboxyl group must be formed in a recombination reaction at the electrode surface in the pH region 2-7, as indicated by the slope of the $E_{1/2}$ -pH line in this interval [11].

Controlled potential electrolysis of (I) in slightly acidic buffers at -1.0 V (SCE) gives the corresponding aldehyde in good yield, provided that the re-

Table 1. Yield of 1-Ethyl-4-Formylpyridinium Bromide from the Cathodic Reduction of 1-Ethyl-4-Carboxypyridinium Bromide as a Function of pH [10] at 25°

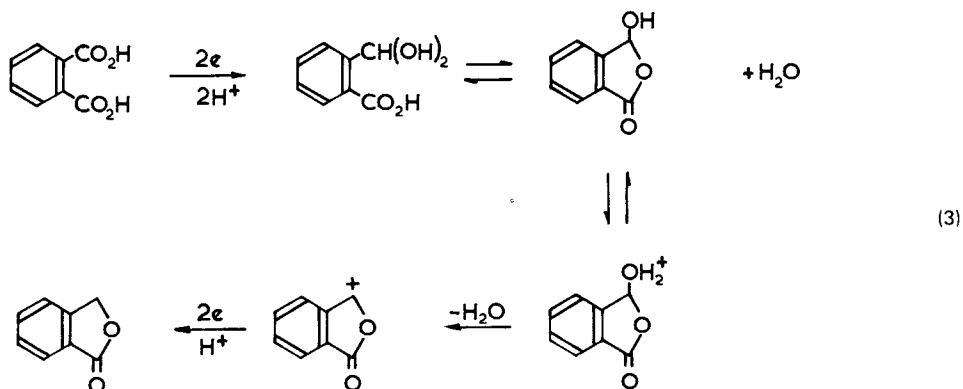
pH	Yield of aldehyde, %
0.15	30
1.35	55
2.55	75
4.45	72
5.15	63
6.29	29
9.15	0

action is stopped at 2 F mol^{-1} (prolonged reaction times led to a further slow reduction of the aldehyde). Table 1 shows that the best yields of aldehyde are obtained in the pH region 2-5. At alkaline pH no aldehyde can be detected, presumably because the heterocyclic ring system becomes the more easily reducible group of the molecule. Aldehyde formation is also negligible at $\text{pH} < 0$, which might be explained by further reduction of the aldehyde to the carbinol or even to the completely reduced product, 1-ethyl-4-methylpyridinium ion. The formation of 2- and 4-picoline by cathodic reduction of pyridine-2- and 4-carboxylic acid, respectively, in strongly acidic solution [12] is a suitable analogy.

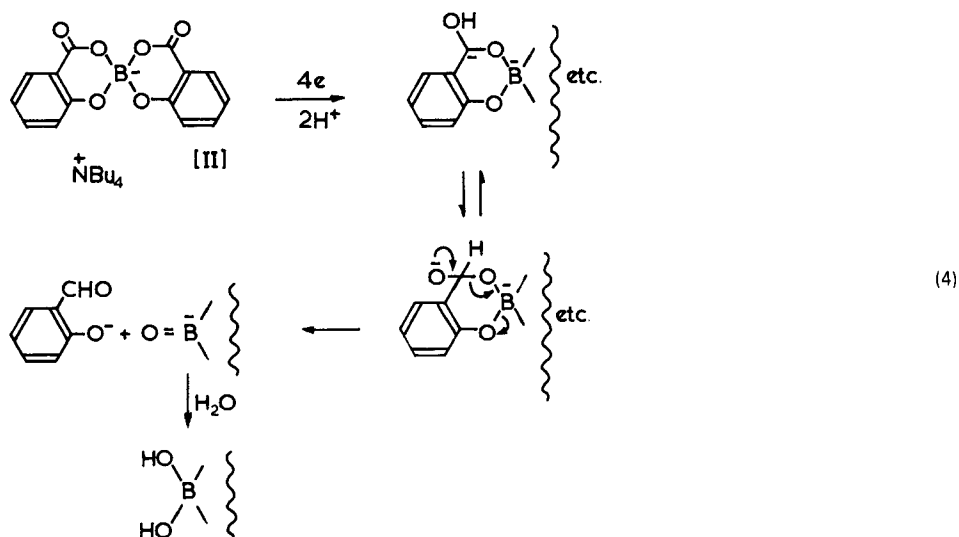
Aldehydes are usually more easily reduced than carboxylic acids but, as in the case above, the aldehyde may exist in aqueous solution predominantly as the hydrate [Eq. (1)]. The hydrates are less easily reduced than carboxylic acids. Several other systems have been found to behave similarly [13,14]: pyridine-2- and 4-carboxylic acid, imidazole-2-carboxylic acid, and thiazole-2-carboxylic acid. In the last case the yield is low due to competing reduction of the ring. Yields are better at low temperatures due to the slowing down of the dehydration reaction.

In summary, it seems [14] that the following requirements must be fulfilled for the reduction of a carboxyl group to an aldehyde function to take place: (1) the carboxyl group must be activated by a strongly electron-withdrawing group; (2) the carboxyl group must be the most easily reducible group in the molecule (which automatically excludes the nitro group and a number of nitrogen-containing heteroaromatic groups as activating groups); and (3) the aldehyde formed must be protected against further reduction by formation of a nonreducible derivative, such as a hydrate or hemiacetal. The last criterion is probably the most important and stringent one.

Other acids which fulfill these requirements and can be reduced cathodically to aldehydes are oxalic acid (to glyoxylic acid) and lactic acid [3]. Phthalic acid, also an activated acid, cannot be reduced to an aldehyde in neutral or weakly acidic electrolytes at a lead cathode but is converted into phthalide, possibly due to the fact that the intermediate phthalaldehydic acid is in equilibrium with hydroxyphthalide [15], which should be easily reducible after protonation and loss of a water molecule [Eq. (3)]:

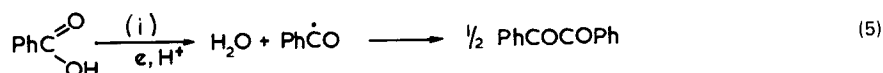


Reduction of phthalic acid in acidic solution (e.g., 5% aqueous H_2SO_4) gives, in good yield, 1,2-dihydrophtalic acid [16-21]; the *trans* and *cis* isomers are formed approximately in the ratio 6:1 [21]. Lead cathodes have been much used for this reduction but during the course of the reaction a porous film of organic detritus forms [22] and mercury would seem to be a better choice as cathode material [23]. Hydration is not the only way in which an aldehyde intermediate may be protected against reduction; the aldehyde may be converted in situ into a borate complex of the hydrate [24-27] or into a bisulfite addition complex [28,29]. In those cases where sodium cations are present the possibility arises of indirect reduction, e.g., via a sodium amalgam. There is compelling evidence that borate complexes, e.g., (II) in the case of salicylic acid reduction, are intermediates. The salt (II) has been isolated and characterized [27] and in the presence of proton donors it undergoes four-electron reduction; a plausible mechanism is suggested in Eq. (4).



In several of the methods referred to above, two-phase systems are used and rapid extraction of aldehyde into an organic phase is important in intercepting the product before further reduction occurs [24-26]. In an important pragmatic study [26] it was shown that correct choice of buffer and a two-phase system allows reductions of aromatic acids to aldehydes in acceptable yield (>50%); in addition the likelihood of such success is greater the lower the $E_{1/2}$ value for reduction of the corresponding methyl ester.

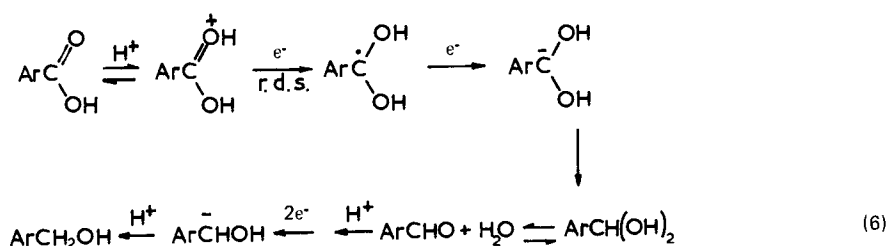
Two other modes of reduction are open to aromatic carboxylic acids. The first is a curiosity; it is reported that benzil is the major product of reduction of benzoic acid at an illuminated germanium cathode [30], a result which is explicable if not understandable in terms of Eq. (5). The other mode of reduction is the easiest to realize, i.e., complete four-electron reduction.



(i) Germanium cathode (p-type); 80% EtOH-H₂O

This reaction is carried out in strongly acid solution, at lead or mercury cathodes, and nonactivated aromatic carboxylic acids give excellent yields of the corresponding benzyl alcohols [18,31-33]. The reaction is easily performed on a large scale, e.g., the adaptation of a filter press cell for continuous reduction of benzoic acid has been described [34].

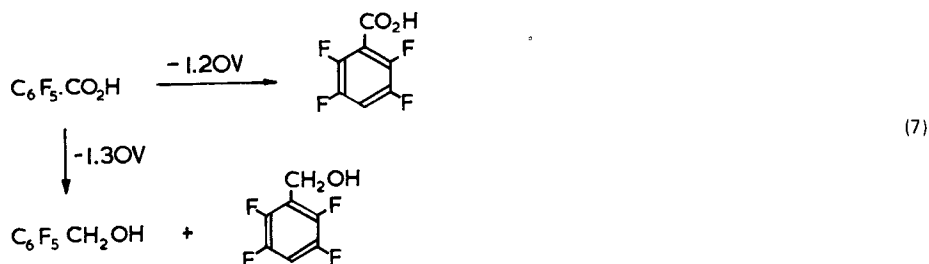
The mechanism of four-electron reduction of benzoic acid has been much studied [35-38]. At mercury the current efficiency (for reduction to a mixture of aldehyde and alcohol) is significantly influenced by the nature of the cation; reduction is more efficient in the presence of tetramethylammonium ions than in the presence of alkali metal ions. This result is interpreted to mean that the important electron-transfer step is to an ion pair in which the benzoate anion's charge is well dispersed [37]. Attempts have also been made to assess the relative importance of the steps most likely to be involved in the four-electron reduction by measuring directly rates of disappearance of aromatic acid or of appearance of product(s) and relating these parameters to substituents effects and potential [35,38]. In both cases the rates of reduction for substituted benzoic acids follow the order predicted by the qualitative theory of substituent effects; in one study [38] a good Hammett plot is obtained. In these systems reduction of the acid has to compete with hydrogen evolution, but bearing in mind that 1 M H₂SO₄ was the electrolyte for the more recent work [38] the mechanism seems best accommodated by Eq. (6);



in more acidic solutions a greater proportion of the carboxylic acid will be protonated. In some cases alcohol is added as a cosolvent and this may also result in the formation of esters which are reduced more easily than the corresponding acid.

Because the carboxylic acid group is so difficult to reduce the method cannot normally be used for acids substituted with easily reducible functions or groups. For instance, pentafluorobenzoic acid loses fluorine by cathodic cleavage at mercury, in 20% sulfuric acid, at -1.20 V (SCE); 2,3,5,6-tetrafluorobenzoic acid is obtained in 75% yield [39]. However, at potentials more

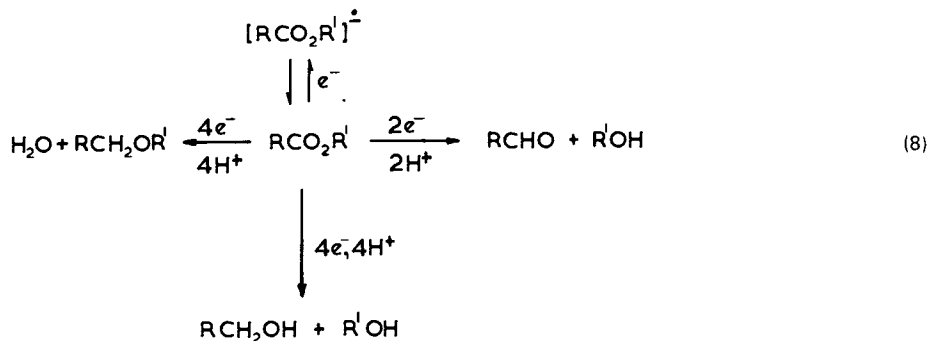
negative than -1.30 V reduction of the carboxylic acid group becomes significant and a mixture of the pentafluoro- and tetrafluorobenzyl alcohols is produced, Eq. (7):



Aliphatic carboxylic acids are reducible, albeit in low yield; strongly acidic electrolytes are required. Thus, phenylacetic acid is reduced to β -phenylethanol in 25-35% yield at a lead cathode in 50% sulfuric acid-ethanol [40], and butyric acid to butanol in 80% sulfuric acid in 6.5% yield [41]. The latter conversion can also be performed in 17% yield in 7% aqueous sodium hydroxide, in spite of the fact that the difficultly reducible carboxylate form of the acid predominates in bulk solution.

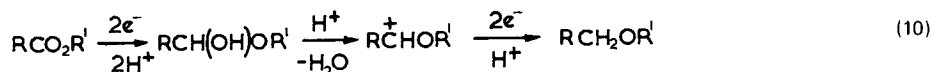
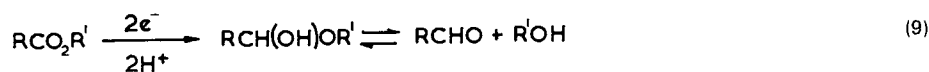
III. CATHODIC REDUCTION OF ESTERS, LACTONES, AND ANHYDRIDES

As with carboxylic acids, nonactivated esters are reduced with difficulty unless strongly acidic solutions are used. Unsaturated esters, e.g., α,β -unsaturated and aromatic esters, are reduced relatively easily because of the activating effect of the strongly electron withdrawing ester group for reduction of the attached unsaturated function. In these cases the products of reduction may include those of hydrodimerization and hydrogenation of the unsaturated function (e.g., aromatic ring) as well as those of reduction of the ester function. The possibilities are summarized in Eq. (8) and subsequent discussion will focus on reduction of the ester function.



The first step in the cathodic reduction of an ester in acidic medium leads to an aldehyde, presumably via the hemiacetal [Eq. (9)]. The aldehyde, if not stabilized in the form of an electroinactive addition compound, is then reduced in a second step to form the corresponding carbinol (or sometimes also the pinacol). However, a competing pathway for ester reduction in

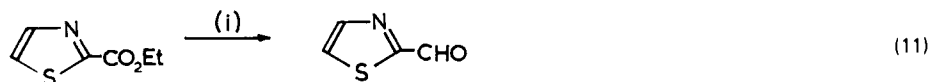
strongly acidic medium is the formation of ethers [Eq. (10)], sometimes in good yields [18,31,42,43].



Voltammetric data for ester reductions are available for several aromatic esters [44-47] and in particular cyclic voltammetry shows clearly that in the absence of proton donors reversible formation of radical-anions occurs [4]. In DMF solution the peak potential for reduction of methyl benzoate is -2.29 V (vs. SCE); for comparison dimethyl terephthalate reduces at -1.68 V and phthalic anhydride at -1.25 V [4]. Half-wave potentials for reduction of Ar-COOCH_3 in an unbuffered solution of pH 7.2 are linearly correlated with σ values [44]; electron-withdrawing substituents in the ring or alkoxy group shift $E_{1/2}$ toward less negative potentials. Generally, esters seem to be more easily reducible than the parent carboxylic acids. Anion radicals of methyl, ethyl, and isopropyl benzoate have been detected by ESR spectroscopy upon cathodic reduction of these esters in acetonitrile/tetrapropylammonium perchlorate [45]. The anion-radicals of several anhydrides, including phthalic anhydride, have similarly been studied [48].

From the preparative point of view, the cathodic reduction of esters, in spite of its potential synthetic usefulness, has been less used than that of carboxylic acids. Nonactivated esters are reduced only slowly, if at all, in strongly acidic medium at a lead cathode [42], whereas methyl or ethyl benzoate is reduced to a mixture of benzyl alcohol and methyl or ethyl benzyl ether, respectively [42,43]. Phenyl and benzyl benzoate give ethers predominantly [18,31], as do also ethyl *o*-chloro- and *m*-bromobenzoate [43].

For activated esters the problem of halting reduction at the aldehyde stage is similar to that faced in the reduction of carboxylic acids; the aldehyde must be converted in situ into an electroinactive form. In one case, that of reduction of ethyl 2-thiazolecarboxylate [14], the strongly hydrated aldehyde may be obtained in about 70% yield [Eq. (11)]. Presumably for similar reasons



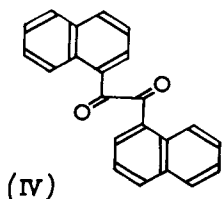
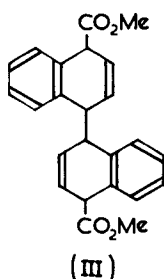
(i) Hg cathode, 0.8N HCl; -0.85 V (vs. SCE).

cathodic reduction of diethyl terephthalate and diethyl furan- and thiophene-2,5-dicarboxylate [44,46] in a buffered ethanolic solution of "pH 4.2" affords the aldehyde ester under CPE conditions (in the interval between -1.50 and -1.65 V). Diethyl oxalate [49] is reduced at a mercury cathode in acidic medium to give the ethyl hemiacetal of ethyl glyoxylate (38-50% yield). On the contrary, diethyl and monoethyl phthalate gave phthalide under similar conditions [16,50].

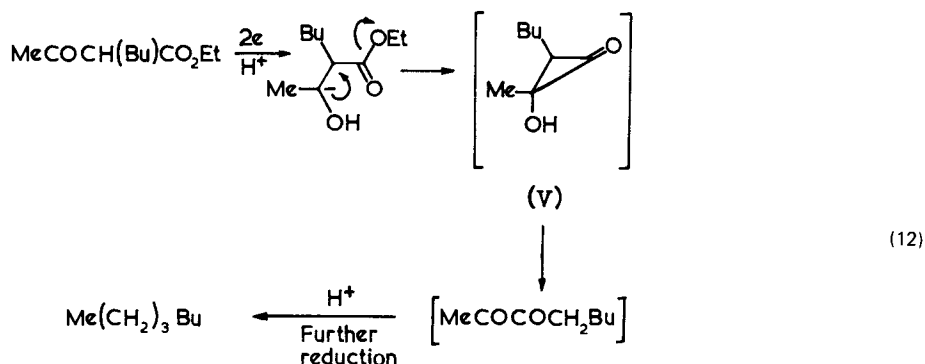
In aprotic conditions diethyl oxalate is reduced reversibly and its relatively stable radical-anion has been well characterized by esr spectroscopy [51,52]. This contradicts an earlier claim that the species formed by one-

electron reduction of diethyl oxalate dimerized with subsequent elimination and decarboxylation to the radical-anion of mesoxalate ($\text{EtO}_2\text{C}\cdot\text{CO}\cdot\text{CO}_2\text{Et}$) [53]. Not all oxalate esters give reversible reduction; the radical-anions of several (e.g., dibenzyl oxalate) undergo rapid fragmentation [54].

Methyl 2-naphthoate, in which the aromatic ring system is also electro-active, is reduced [55] in the presence of boric acid in methanolic solution at a mercury cathode to form 2-naphthalenemethanol in 19% yield, a mixture of diastereomeric pinacols from further reduction of the intermediate aldehyde in 19% yield, and a mixture of methyl 1,2- and 1,4-dihydro-2-naphthoates in 42% yield. The tendency for ring attack is still more pronounced in the case of methyl 1-naphthoate which under the same conditions [56] gives methyl 1,4-dihydro-1-naphthoate (43%) dimer (III) (20%), di-1-naphthylglyoxal (IV) (5%), and 1-naphthalenemethanol (5%). Because mercury cathodes are used with sodium salt electrolytes it is possible that electrogenerated sodium amalgam is the reducing agent.

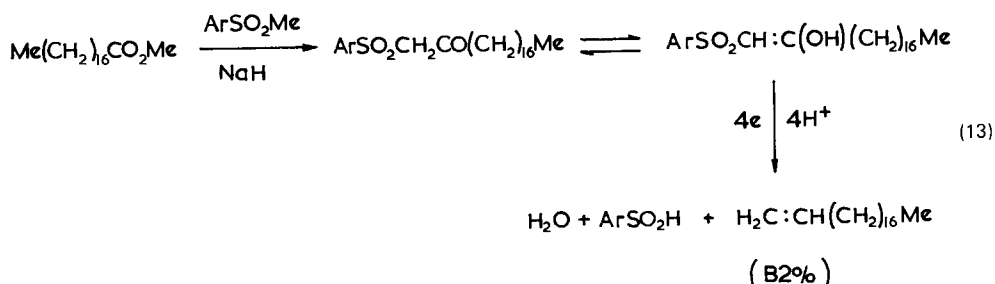


2-Alkyl-substituted acetoacetic esters undergo an interesting rearrangement reaction (the Tafel rearrangement) upon reduction at a lead cathode in an acidic medium [Eq. (12)] with complete reduction of the functional groups [57-62]. The cathodic reduction of β -diketones gives analogous rearrange-

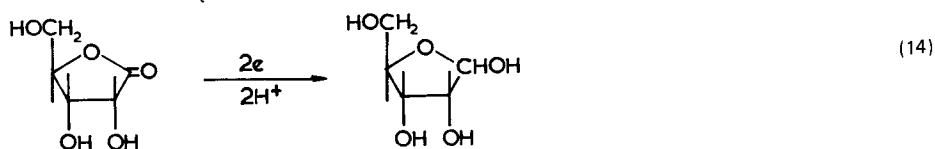


ments and, because these are known to involve cyclopropanediol intermediates [63], it is tempting to suggest that the series (V) is involved in the Tafel rearrangement. In partial support of Eq. (12) there is evidence, although not by isolation, that 2,3-octanedione is one of the many by-products of the Tafel rearrangement of ethyl 2-butylacetoacetate [62].

Two essentially indirect reductions of nonactivated esters are worth reporting. Dimethyl adipate is reduced in HMPA by electrogenerated solvated electrons; the sole product is *n*-hexane [64]. Terminal alkenes may be prepared electrochemically from esters (and lactones) provided that the ester is first converted into a β -hydroxysulfone, which reduces cathodically with concomitant 1,2-elimination [65] [Eq. (13)].



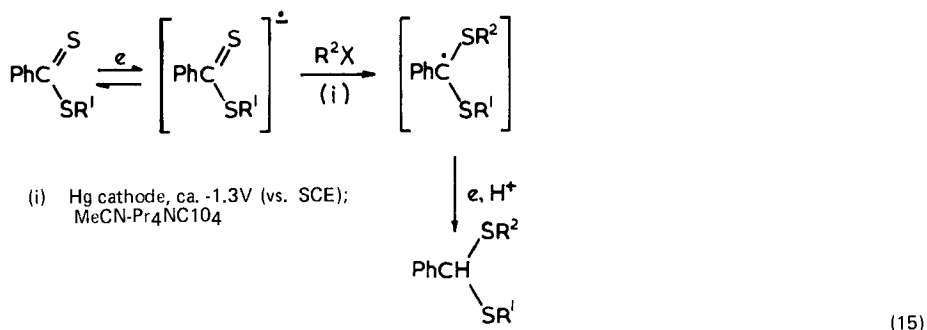
Lactones are in principle reducible to hydroxyaldehydes and when applied to lactones of sugar acids the method works well because cyclic acetal or ketal formation stabilizes the carbonyl function and prevents overreduction [3,66-70]. In particular much attention has been paid to the production of D-ribose from the cathodic reduction of D-ribono- γ -lactone [Eq. (14)]; the optimum conditions for this reaction include careful buffering of the electrolyte [67-70]. The reaction most probably proceeds by electrochemical formation of an



amalgam which chemically reduces the lactone; mercury cathodes are favored but it has been shown that lead cathodes work in the presence of lithium or sodium cations provided that the rate of regeneration of the lead-alkali metal alloy is high and not inhibited unduly by adsorption of the lactone [71].

Thiol esters of aromatic acids, ArC(O)SR , are cathodically reduced with fission of the S-CO bond, affording the thiol in good yield [72], e.g., L-cysteine in 83% yield from S-benzoyl-L-cysteine at a mercury cathode in DMF-water-tetramethylammonium chloride. The electroreduction of many other thioesters [containing $-\text{C(O)SR}$, $-\text{C(S)OR}$, and $-\text{C(S)SR}$ functions] has been systematically studied; in many cases stable radical anions are produced under aprotic conditions [52,73,74]. Such radical ions are often reactive toward added electrophiles such as alkyl halides and the reactions outlined in Eq. (15) have been realized in moderate yields (25-50%) [75].

Anhydrides have received very little attention as substrates for cathodic reduction. Polarography of cyclic anhydrides [76] in acetonitrile and DMF displayed two one-electron waves for the most easily reduced anhydrides

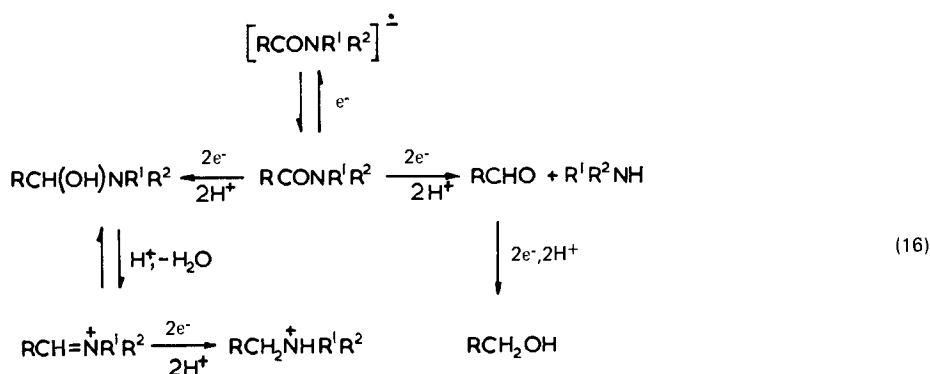


(pyromellitic, dibromopyromellitic, and tetrachlorophthalic anhydride). For 10 mono- and dianhydrides of aromatic carboxylic acids the $E_{1/2}$ values for reduction in acetonitrile have been measured and listed [77].

Controlled potential electrolyses of phthalic and pyromellitic anhydrides at the potential of the first wave gave highly colored, oxygen-sensitive solutions, which later were shown to give well-resolved esr spectra of the corresponding radical ions [48]. Phthalic anhydride has been reduced to phthalide on a preparative scale [78], but the reaction conditions employed (ethanol-water-ammonium carbonate) indicate that monoethyl phthalate must have been the actual substrate.

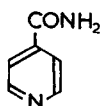
IV. CATHODIC REDUCTION OF AMIDES, LACTAMS, IMIDES, AND HYDRAZIDES

The carbonyl group of amides, lactams, and imides is reducible at lead and mercury cathodes in strongly acidic media, usually to the corresponding methylene compound. In media of lesser acidity cathodic cleavage may occur and, depending on the structure of the starting material and the detailed reaction conditions, the products outlined in Eq. (16) are accessible by cathodic reduction.



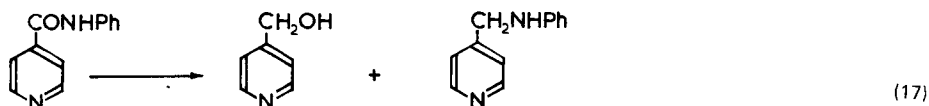
Voltammetric and esr spectroscopic studies, using aprotic conditions, show that several amides give reversible one-electron reduction to relatively stable radical anions, e.g., for N,N-dimethylbenzamide [4], and N-alkylphthalimides [48,79-81] and oxamides [82]. In a similar fashion the electrogeneration of radical anions from sulfur analogs [containing the $\text{RC(S)NR}^1\text{R}^2$, $\text{RR}^1\text{NC(S)C(S)NR}^2\text{R}^3$, and $\text{ArC(S)CONR}^1\text{R}^2$ functions] has been thoroughly investigated [83-85].

The polarography of pyridine-4-carboxamide and some of its derivatives [86], as well as carboxamides of other heterocyclic systems [13,14], has been carefully studied in aqueous medium over the whole pH range. As an example, it is instructive to see how structural variations in and at the amide group of pyridine-4-carboxamide change the product distribution. Pyridine-4-carboxamide (VI) displays a polarographic wave with $E_{1/2}$ ranging from about -0.6 V at pH 0 to -1.3 V at pH 13, the change being related to the extent of protonation. Controlled potential electrolysis of (VI) in 0.8 M hydrochloric acid



(VI)

at -0.80 V (SCE) gave a good yield of pyridine-4-aldehyde, whereas at pH 4.5, 4-pyridinemethanol was formed. The latter reaction is somewhat difficult to explain since one would expect the aldehyde to be significantly hydrated at this pH. One possibility is that the acetate buffer used in this experiment has a strong catalytic effect on the dehydration process. Substitution of hydrogen by phenyl in the amide group changes the product composition considerably; electrolysis of N-phenyl-4-pyridinecarboxamide in 1 M hydrochloric acid at -0.75 V gives a mixture of 4-pyridinemethanol and 4-anilinomethylpyridine [Eq. (17)]:



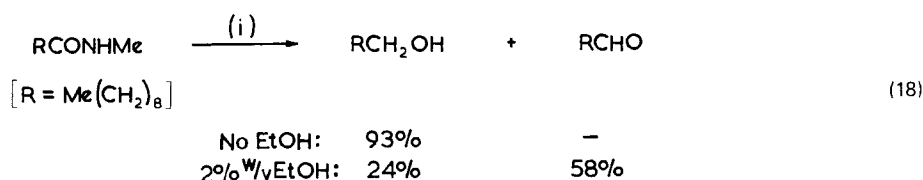
Pyridine-4-thiocarboxamide is reduced in two two-electron steps in acidic medium; at the potential of the second wave reduction gives 4-aminomethylpyridine. These three cases show that the preparative outcome of an amide reduction is strongly dependent on the substituents at the amide group and that the changes manifest themselves in the chemical step [Eq. (16)]. For amides of aromatic carboxylic acids the reduction potential, if not the outcome of reaction, may be gleaned from a comprehensive compilation [87].

A large number of preparative applications of amide group reduction have been compiled and discussed by other authors [1-5]. Generally, these reactions are performed at lead cathodes in aqueous sulfuric acid or ethanolic aqueous sulfuric acid and result in good yields of the corresponding amino-methyl compounds. Since the media employed are so strongly acidic it is probable that the electroactive species is the protonated amide in these cases. Simple aliphatic amides of the general formula $RCONH_2$ have not been studied; however, a number of N,N-dimethyl- or N,N-diphenyl-substituted aliphatic amides have been successfully reduced. Benzamides give good yields of benzylamines, N,N-disubstitution favoring amine formation also in these cases. Thioamides can be used as substrates with similar results.

In neutral or alkaline solution, N-substituted benzamides undergo cathodic cleavage to form benzyl alcohol and an amine [87,88]. The reaction is performed in an initially neutral solution of tetramethylammonium chloride in

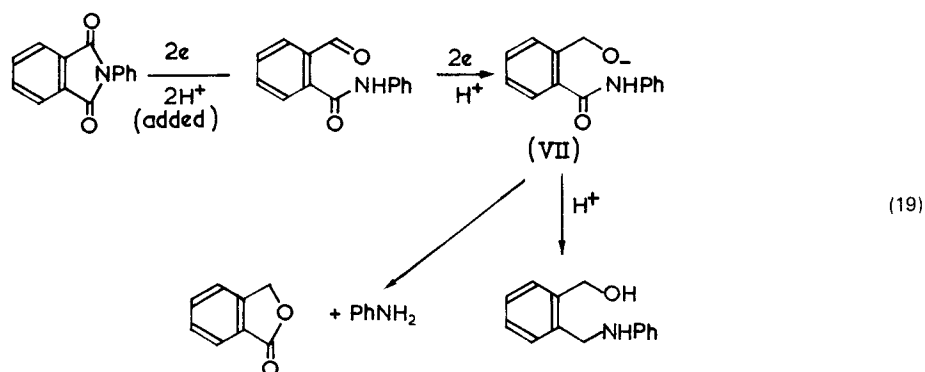
methanol, but in a divided cell the catholyte quickly becomes alkaline, so the reaction proceeds under alkaline conditions during most of the electrolysis period.

In amine solvents the reduction of amides by electrogenerated solvated electrons has some preparative value [89]. Platinum electrodes are used with a lithium chloride-methylamine electrolyte and the ratio of alcohol to aldehyde product depends on the presence or otherwise of a proton donor [Eq. (18)].



(i) Pt electrodes; LiCl-MeNH₂; -5°.

Phthalimide and substituted phthalimides are relatively easily reduced [*E*_p/2 in DMF ca. -1.4 V (vs. SCE)]. In alkaline or aprotic solution two one-electron reduction steps are observed in voltammetric experiments [90-93]; anion radical formation at the first wave has been confirmed by esr spectroscopy [48,91,92]. Considerable attention has been given to the detailed mechanism of reduction of phthalimide derivatives [90-93]. The combined results of voltammetric and product-oriented studies suggest strongly for nonaqueous solution the mechanism in Eq. (19). Particularly relevant is



the formation of O-acetyl-2-hydroxymethylbenzanilide as a result of reduction [92] of N-phenylphthalimide in DMF in the presence of acetic anhydride; the result is well rationalized as being due to the trapping intermediate (VII), Eq. (19). In acidic solution, and in the presence of strong proton donors in nonaqueous solvents, a single two-electron reduction wave is observed [93]. In macroscale electrolyses further reduction takes place easily, lactams or fully saturated cyclic amines being obtainable in good yields from imides in strongly acidic electrolytes [1-5].

The reduction of acyl hydrazides has been investigated very little. Polarography of 4-pyridinecarboxylic acid hydrazide in acid and neutral solution shows that it is reduced in two two-electron steps [94]. The result of controlled potential electrolysis is cleavage of the N—N bond to form 4-pyridine-

carboxamide, which then undergoes reduction to the aldehyde in the second step. Other hydrazides of heterocyclic carboxylic acids behave analogously.

V. CATHODIC REDUCTION OF NITRILES

Both aliphatic and aromatic nitriles are reduced in acidic solution to give amines [Eq. (20)], whereas in neutral or alkaline medium cleavage of the C—CN bond takes place [Eq. (21)]:



α, β -Unsaturated nitriles, however, undergo ready hydrodimerization under all but acidic conditions (Chap. 20).

Reactions (20) and (21) have been studied polarographically [95]. In acid solution, aromatic nitriles of the type $p\text{-RCOC}_6\text{H}_4\text{CN}$ ($\text{R} = \text{H}$ or CH_3) undergo a four-electron reduction at -0.70 V (SCE) with formation of the corresponding benzylamine. The protonated nitrile is the electroactive species. In neutral medium, nitriles of the type $\text{XC}_6\text{H}_4\text{CN}$ ($\text{X} = p\text{-COO}^-$, $p\text{-SO}_2\text{NH}_2$, and $m\text{-SO}_2\text{NH}_2$) give two-electron reduction with formation of cyanide ion and $\text{C}_6\text{H}_5\text{X}$. This duality in mechanism has been thoroughly studied in the case of 2- and 4-cyanopyridine [96,97], and it was shown that both pH and electrode material play an important role in determining the mechanism. The electrochemical cleavage of nitriles can also be performed in anhydrous amine media [98], making possible C—CN bond cleavage in aliphatic nitriles such as cycloheptanecarbonitrile and heptanecarbonitrile. This reaction was interpreted as being due to reductive cleavage by solvated electrons (Chap. 27).

A number of aromatic nitriles including benzonitrile have been reduced electrochemically in DMF-tetrapropylammonium perchlorate to give solutions of relatively stable anion radicals, whose ESR spectra have been recorded [99]. Phthalonitrile, which gives two one-electron waves on polarography in this medium, undergoes decyanation on CPE at the potential of the second wave.

The preparative use of nitrile reduction has been primarily for the synthesis of amines [1,2,100-104]. This is done in acidic medium and gives fair to good yields. Electrocatalytic hydrogenation of nitriles is also sometimes efficient [105]; in these cases nickel or palladium black, deposited on the cathode, promote reduction by electrolytically generated hydrogen.

Imino ether hydrochlorides can be reduced to amines in good yields at lead cathodes in aqueous sulfuric acid [106]. Cathodic reduction of nitriles in ethanolic sulfuric acid should be a useful procedure therefore, with the first step being in situ formation of an imino ether salt.

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